APPLIED STATISTICS, MATH10096

Workshop 5

1. A blood pressure drug trial looked at change in blood pressure, y_i, over 6 weeks of therapy for 40-45 year old hypertensive patients. Patients were given one of 3 drug formulations or a placebo, each at one of three dose rates. The data for 120 patients were modelled as

$$\mathbb{E}(y_i) = \mu + \alpha_j + \beta_k, \text{ if patient } i \text{ is formulation } j, \text{ dose group } k,$$

with the y_i assumed to be independent gamma random variables. Model checking plots suggested that the full model assumptions were met, and the model was re-fitted without the α_j terms. The deviance for the full model fit was 193, while the deviance for the reduced version was 201. What hypothesis can you test using this information, what is the associated p-value, and what can you conclude?

2. If you add a completely irrelevant predictor variable, with one associated parameter, to a GLM, how much decrease do you expect in the scaled deviance of the GLM?

3. This question covers the Belgian AIDS epidemic data discussed in Section 10.1 of the lecture notes. The data can be entered into R as follows.

```
y <- c(12,14,33,50,67,74,123,141,165,204,253,246,240)
t <- 1:13
```

Recall that y refers to the number of new AIDS cases per year and t denotes the year since 1980 (up to and including 1993).

(a) Plot AIDS cases against year since 1980.

(b) Use the `glm` function to fit the model for the AIDS data given in the notes. You will need to use the `poisson` family and `"log"` link function.

(c) Based on the scaled deviance, does the model appear to be adequate?

(d) Examine the residuals for the fitted model. Is the model adequate?

(e) Propose an alternative model for the data and fit it.

(f) Check the alternative model, and see if all its terms are 'significant', using the `summary` command.

(g) Use AIC to compare your revised model and the original model.

(h) Which model do you think best describes the data?

(i) What can you conclude about whether or not the epidemic is continuing to increase exponentially by the end of the data series?

4. This question is about analyzing the O-rings data discussed in Question 4 of workshop sheet 4. The data is available on Learn in the file `orings.rda` and can be read in `R` as follows.

```r
load("orings.rda")
```

Recall the context: in January 1986 the space shuttle Challenger exploded shortly after take-off. Subsequent investigation eventually focused on the possibility of o-rings in the fuel tanks having failed as a result of the unusually low launch temperature (31 degrees F), hence allowing a fuel leak. You can read more in Wikipedia:

https://en.wikipedia.org/wiki/Space_Shuttle_Challenger_disaster

It turned out that data on o-ring failure in previous launches was available for various launch temperatures. In particular, for each temperature considered, we have the number of o-rings, out of 6, that failed/were damaged. As discussed in the solutions of workshop 4, a logistic regression model might be appropriate. Viewing each of the 6 o-rings at each launch temperature as independent binomial trials, with a probability of failure that depends on temperature, gives:

$$\operatorname{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_1 + \beta_2 \texttt{temp}_i, \quad y_i \overset{\text{indep.}}{\sim} \operatorname{binom}(p_i, 6), \tag{1}$$

where p_i denotes the probability of o-ring failure/damage at temperature temp_i and y_i denotes the number of damaged o-rings, out of 6, at temperature temp_i.

(a) Reproduce the plot from workshop 4 sheet.

(b) Fit the model suggested in (1) in `R`. You will need to use the `binomial` family and `"logit"` link function.

(c) The standard residual plots for this model are of little use. Why is that?

(d) To get a feel for the plausibility of the model, overlay a curve showing the expected number of failures as a function of temperature on your original data plot. Plot also approximate 95% confidence limits for the mean number of failures at each temperature. Note that an inverse function of the logit link can be created in `R` as follows:

```r
ilogit <- binomial()$linkinv
```

(e) Obtain a point estimate and approximate 95% CI for the expected number of o-ring failures at the Challenger launch temperature of 31 degrees F. What does this suggest? How reliable are the results likely to be?

(f) In some respects the first observation in the `orings` dataset looks out of line with the others, and is an extremely influential point in the fit. Use the `cooks.distance` function applied to your fitted model object to confirm this. What do you think should be done about this?

5. In this question, we will analyze the data in the ‘contingency’ table from Question 3 in workshop 4 exercise sheet. Contingency tables occur when you have observations of several factor variables for a number of study subjects, and what is of interest is how many subjects/observations occur at each combination of the factor variables. The questions of interest are usually about whether the factors are linked in some way. Question 3 on workshop sheet 4 concerns a simple example of a contingency table, useful for investigating whether there is an association between gender and belief in life after death. We can address this question by using an analysis of deviance to compare the fit of two competing models of these data: one in which belief is modelled as independent of gender, and another in

which there is an interaction between belief and gender. The data are as follows:

	Believer	Non-Believer
Female	435	147
Male	375	134

The data can be entered in R as follows (below 1 stands for believer and 0 for non-believer).

```
al <- data.frame(y = c(435, 147, 375, 134),
                 gender = as.factor(c("F","F","M","M")),
                 belief = as.factor(c(1, 0, 1, 0)))
```

As we have seen, under the null model of independence between belief and gender, and if y_i is an observation of the counts in one of the cells of the table, then the expected count can be modeled as

$$\mu_i \equiv \mathbb{E}(Y_i) = n\gamma_k\alpha_j \text{ if } y_i \text{ is data for gender } k, \text{ and faith } j.$$

where n is the total number of people sampled, γ_1 and γ_2 are the proportion female and male in the population sampled, and α_1 and α_2 are the proportion of believers and non-believers in the population sampled. As seen in the previous workshop, we can yield a linear predictor by taking logs

$$\log(\mu_i) = \log(n) + \log(\gamma_k) + \log(\alpha_j) \text{ if } y_i \text{ is gender } k \text{ and faith } j,$$

and defining $\tau = \log(n)$, $\beta_k = \log(\gamma_k)$ and $\delta_j = \log(\alpha_j)$, we get

$$\log(\mu_i) = \tau + \beta_k + \delta_j \text{ if } y_i \text{ is gender } k \text{ and faith } j.$$

Note that the appropriate distribution for the data is technically multinomial. However, the resulting likelihood is equivalent to the likelihood we would obtain by assuming that the counts in each cell of the table are observations from Poisson random variables. Therefore, the Poisson distribution can be used in practice.

(a) Fit the model assuming that belief and gender are independent. Since there are only 4 residuals with 1 degree of freedom between them, there is no point looking at residual plots for this one. Based on the scaled deviance, does the model appear to be adequate?

(b) Do these data provide any evidence for an association between gender and belief in an afterlife?

6. This exercise involves data from a dose-response study. Specifically, it pertains to the number of adult flour beetles that died after five hours of exposure to gaseous carbon disulfide at various dosages. The data can be entered into R as follows

```
logdose <- c(1.691, 1.724, 1.755, 1.784, 1.811, 1.837, 1.861, 1.884)
dead <- c(6, 13, 18, 28, 52, 53, 61, 60)
n <- c(59, 60, 62, 56, 63, 59, 62, 60)
```

The goal of this exercise is to fit a binomial regression model using three link functions and assess which one provides a better fit to the data. Let p_i be the probability of death at (log) dose level x_i. The three link functions to be considered are:

- *Logit link function*:

$$\log\left(\frac{p_i}{1-p_i}\right) = \eta_i = \beta_0 + \beta_1 x_i.$$

- *Probit link function*:

$$\Phi^{-1}(p_i) = \eta_i = \beta_0 + \beta_1 x_i,$$

  where Φ stands for the standard normal cumulative distribution function. In `R`, please specify `family = binomial(link = "probit")`.

- *Complementary log-log link function*:

$$\log(-\log(1-p_i)) = \eta_i = \beta_0 + \beta_1 x_i.$$

  In `R`, please specify `family = binomial(link = "cloglog")`.

7. The dataset `africa` from `R` package `faraway` gives information about the number of military coups in sub-Saharan Africa and various political and geographical information. Namely, the following variables are available:

   - `miltcoup`: number of successful military coups from independence to 1989.
   - `oligarchy`: number years country ruled by military oligarchy from independence to 1989.
   - `pollib`: Political liberalization - 0 = no civil rights for political expression, 1 = limited civil rights for expression but right to form political parties, 2 = full civil rights.
   - `parties`: Number of legal political parties in 1993.
   - `pctvote`: Percent voting in last election.
   - `popn`: Population in millions in 1989.
   - `size`: Area in 1000 square km.
   - `numelec`: Total number of legislative and presidential elections.
   - `numregim`: Number of regime types.

   The below `R` session attempts to build a simple model of the number of military coups, based on the other variables.

   (a) Write short notes to accompany the `R` session, explaining what is being done, statistically, and what model you would finally choose for these data.

   (b) Interpret the coefficients of the finally selected model. Do they make sense?

   (c) What is the 'proportion deviance explained' for the selected model?

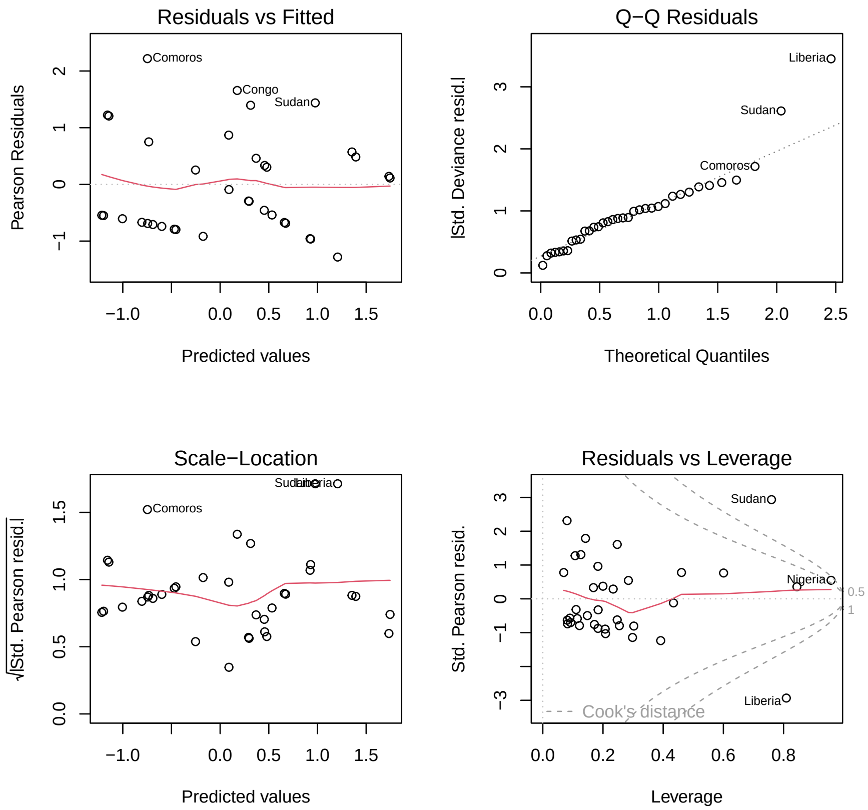
   (d) Is the model a useful predictive model for number of coups?

   (e) From the models fitted which model would you have selected if using AIC for model selection?

```
library(faraway)
data(africa)
ac1 <- glm(miltcoup ~ oligarchy + as.factor(pollib) +parties +
             pctvote + popn + size + numelec + numregim,
```

```r
            family = poisson, data = africa)

par(mfrow = c(2,2))
plot(ac1)
```




```r
drop1(ac1, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##     popn + size + numelec + numregim
##                   Df Deviance    AIC    LRT Pr(>Chi)
## <none>                 28.249 113.06
## oligarchy          1   32.354 115.16 4.1045 0.042769 *
## as.factor(pollib)  2   35.581 116.39 7.3314 0.025587 *
## parties            1   35.311 118.12 7.0616 0.007875 **
## pctvote            1   30.572 113.38 2.3222 0.127537
## popn               1   30.601 113.41 2.3513 0.125178
## size               1   29.238 112.05 0.9881 0.320197
## numelec            1   28.430 111.24 0.1810 0.670499
## numregim           1   29.059 111.87 0.8099 0.368147
```

```r
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac2 <- update(ac1, .~. - numelec)
drop1(ac2, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##     popn + size + numregim
##                   Df Deviance    AIC    LRT Pr(>Chi)
## <none>                 28.430 111.24
## oligarchy          1   36.595 117.40 8.1649 0.004271 **
## as.factor(pollib)  2   36.872 115.68 8.4417 0.014686 *
## parties            1   35.773 116.58 7.3428 0.006733 **
## pctvote            1   30.590 111.40 2.1600 0.141648
## popn               1   30.605 111.41 2.1746 0.140305
## size               1   29.452 110.26 1.0214 0.312196
## numregim           1   29.081 109.89 0.6508 0.419818
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac3 <- update(ac2, .~. - numregim)
drop1(ac3, test = "Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##     popn + size
##                   Df Deviance    AIC     LRT  Pr(>Chi)
## <none>                 29.081 109.89
## oligarchy          1   40.291 119.10 11.2094 0.0008139 ***
## as.factor(pollib)  2   37.830 114.64  8.7487 0.0125960 *
## parties            1   36.304 115.11  7.2224 0.0071998 **
## pctvote            1   31.599 110.41  2.5175 0.1125903
## popn               1   30.614 109.42  1.5324 0.2157552
## size               1   30.040 108.85  0.9582 0.3276300
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac4 <- update(ac3, .~. - size)
drop1(ac4, test = "Chisq")

## Single term deletions
##
## Model:
```

```
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote +
##     popn
##                   Df Deviance    AIC     LRT Pr(>Chi)
## <none>                 30.040 108.85
## oligarchy          1   40.468 117.28 10.4283 0.001241 **
## as.factor(pollib)  2   38.022 112.83  7.9826 0.018476 *
## parties            1   37.547 114.36  7.5070 0.006146 **
## pctvote            1   32.241 109.05  2.2010 0.137920
## popn               1   31.069 107.88  1.0299 0.310189
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac5 <- update(ac4, .~. - popn)
drop1(ac5,test="Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties + pctvote
##                   Df Deviance    AIC     LRT  Pr(>Chi)
## <none>                 31.069 107.88
## oligarchy          1   48.196 123.00 17.1266 3.497e-05 ***
## as.factor(pollib)  2   39.762 112.57  8.6926   0.01295 *
## parties            1   37.547 112.36  6.4773   0.01093 *
## pctvote            1   32.822 107.63  1.7521   0.18561
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac6 <- update(ac5,.~. - pctvote)
drop1(ac6,test="Chisq")

## Single term deletions
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib) + parties
##                   Df Deviance    AIC     LRT  Pr(>Chi)
## <none>                 42.235 125.92
## oligarchy          1   64.574 146.25 22.3389 2.285e-06 ***
## as.factor(pollib)  2   48.597 128.28  6.3615   0.04155 *
## parties            1   45.431 127.11  3.1964   0.07380 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

ac7 <- update(ac6, .~. - parties)
drop1(ac7, test = "Chisq")

## Single term deletions
```

```
##
## Model:
## miltcoup ~ oligarchy + as.factor(pollib)
##                    Df Deviance    AIC     LRT  Pr(>Chi)
## <none>                  45.431 127.11
## oligarchy           1   70.453 150.13 25.0219 5.668e-07 ***
## as.factor(pollib)  2   50.101 127.78  4.6694   0.09684 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(ac7)
```

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + as.factor(pollib), family = poisson,
##     data = africa)
##
## Coefficients:
##                     Estimate Std. Error z value Pr(>|z|)
## (Intercept)           0.4795     0.4274   1.122   0.2618
## oligarchy             0.1023     0.0207   4.942 7.73e-07 ***
## as.factor(pollib)1  -0.5393     0.4695  -1.149   0.2506
## as.factor(pollib)2  -0.9168     0.4426  -2.071   0.0383 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 79.124  on 41  degrees of freedom
## Residual deviance: 45.431  on 38  degrees of freedom
##   (5 observations deleted due to missingness)
## AIC: 127.11
##
## Number of Fisher Scoring iterations: 5
```

8. Assume that Y follows a negative binomial distribution with probability mass function given by

$$f(y;\mu,k)=\frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)}\left(\frac{\mu}{\mu+k}\right)^{y}\left(\frac{k}{\mu+k}\right)^{k},\quad y=0,1,2,\ldots$$

Further assume that $k>0$ is a given constant and consider the random variable $Y^*=Y/k$. Then $\Pr(Y^*=y^*)=\Pr(Y=ky^*)$ such that Y^* has probability mass function given by

$$f(y^*;\mu,k)=\frac{\Gamma(ky^*+k)}{\Gamma(k)\Gamma(ky^*+1)}\left(\frac{\mu}{\mu+k}\right)^{ky^*}\left(\frac{k}{\mu+k}\right)^{k},\quad y^*=0,\frac{1}{k},\frac{2}{k},\ldots \tag{2}$$

(a) Show that (2) is a distribution in the exponential family with

$$\theta=\log\left(\frac{\mu}{\mu+k}\right),\quad b(\theta)=-\log(1-e^{\theta}),\quad a(\phi)=\frac{1}{k}.$$

(b) Find the mean and variance of Y^* using the expressions for $b(\theta)$ and $a(\phi)$. Use these results to show that $\mathbb{E}(Y) = \mu$ and $\text{var}(Y) = \mu + \frac{\mu^2}{k}$.

(c) Compare the relationship between the mean and variance in the negative binomial distribution to that in the Poisson distribution, and comment on when the Poisson model is appropriate and when the negative binomial is needed.